Pragmatic techniques for defending materialism

Topics in Mind and Language: Conditionals

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1. The material conditional theory of indicative conditionals

An indicative conditional of the form 'If ϕ, ψ' is true at all and only those possible worlds where ψ is true or ϕ is not true.

- We can get this as a limiting case from the variably strict theory by taking the accessibility relation to be one that every world bears to itself and no other world.

2. Reasons for thinking 'If P, Q' entails 'P ⊃ Q'

This follows (given classical logic for 'or' and 'not') from the validity of Modus Ponens.

1.	$P > Q, P \vdash Q$	MP
2.	$Q \vdash \neg P \lor Q$	$\lor$-intro
3.	$P > Q, P \vdash \neg P \lor Q$	1,2
4.	$P > Q, \neg P \vdash \neg P$	reiteration
5.	$\neg P \vdash \neg P \lor Q$	$\lor$-intro
6.	$P > Q, \neg P \vdash \neg P \lor Q$	4,5
7	$P > Q \vdash \neg P \lor Q$	3,6, $\lor$-elim

3. Arguments for the material conditional theory

The Or-to-if inference: 'P or Q; therefore if not-P then Q'.

- Seems valid! 'It's either a dog or a cat; so if it's not a cat it's a dog'.
- Or-to-if is valid if the indicative conditional is like ⊃ in classical logic in obeying the metarule of *conditional proof*: whenever $P_1,\ldots,P_n \vdash Q$, $P_1,\ldots,P_{n-1} \vdash P_n > Q$

$P \lor Q, \neg P \vdash Q$	(disjunctive syllogism)
$P \lor Q \vdash \neg P > Q$	(conditional proof)

- Gibbard's proof: Or-to-if can be derived from uncontroversial principles of conditional logic together with the *Import-export* principle, according to which 'If A, then if B then C' is equivalent to 'If A and B, then C'. (Actually we only need the right-to-left, 'import' direction of this.).

$\vdash ((P \lor Q) \land \neg P) > ((P \lor Q) \land \neg P)$	(ID)
$\vdash ((P \lor Q) \land \neg P) > Q$	Weakening the consequent
$\vdash (P \lor Q) > (\neg P > Q)$	(Import-Export)
$P \lor Q \vdash \neg P > Q$	(MP)

A worry about this argument: Import-Export feels valid for counterfactuals too, but Or-to-if seems ridiculous for them! Hmm.

4. A challenge to the material conditional theory:

Objection: asserting 'If P, Q' in a situation where you just know not-P, or just know Q, seems very bad and misleading. But according to the material conditional theory, you know it's true in this kind of case.

5. An analogy

According to orthodoxy, 'P or Q' is entailed by P and entailed by Q. One might object to the orthodoxy on the grounds that when you just know one of P and Q and have no evidence about the other, asserting 'P or Q' would be very bad and misleading, even though according to orthodoxy you're in a position to know that it's true.

Misleading how? Saying 'P or Q' will generally lead hearers to conclude that you don't know either way.

Response: a sentence can be misleading and bad to assert even if it's true.

6. Grice on conversational implicature

Picture: people generally expect one another to be not only literally truthful, but also cooperative. Hearers thus can be expected to make many inferences over and above believing the literal content of the sentence. Speakers can anticipate this and exploit it to convey additional information.

- 'His handwriting is good and he is always on time to class'

Grice offers some "Maxims" which capture what's involved in being cooperative (in the way we expect). They can all be used to generate implicatures: from the fact that someone says this instead of that, we will draw the conclusion that saying that would

> Maxim of Quality. Make your contribution true; so do not convey what you believe false or unjustified.
>
> Maxim of Quantity. Be as informative as required.
>
> Maxim of Relation. Be relevant.
>
> Maxim of Manner. Be perspicuous; so avoid obscurity and ambiguity, and strive for brevity and order.

In the case of 'or', the key maxims are those of quality, quantity and relation. 'P' and 'Q' will almost always be relevant whenever 'P or Q' is, and they are more informative (if true). So from the fact that someone says 'P or Q' rather than 'P' or 'Q' we will generally conclude that they don't believe P and don't believe Q.

- It's a bit complicated. E.g. maybe they actually do believe P but not on the basis of evidence that they think you'd be convinced by, and that's why they are saying only 'P or Q'. Or maybe the obstacle is one of knowledge rather than belief, or lack of high probability — 'Either I'll win the lottery or I won't be able to afford to go on a Safari next year'.
- Thus it's not as simple Jackson's 'Assert the stronger (when probabilities are close)'. [the stronger of what?]

A key feature of converstional implicatures: cancelability.

7. Can conversational implicature save the material conditional theory?

Yes, according to Grice.

Objections from Jackson:

1. It can be OK to assert 'If P, Q' even when one is pretty sure that ¬P.
2. It can be OK to assert 'If P, Q' even when one is pretty sure that Q.
3. Given materialism, 'Either (If P, Q) or (If Q, R)' is a logical truth. Hard to find a pragmatic theory that explains why things of this form are so often bad to assert without having the obviously wrong consequence that logical truths are never good to assert.
4. Sentences that are logically equivalent according to the materialist are assertible under very different conditions in a way that Grice's theory can't explain. Jackson's example: 'Not P and if P, Q' and 'Not P and if P, R'.
5. An argument from 'classroom experience'!?

(1) and (2) don't look problematic given how Grice is really thinking — they are anyway not problematic for the view that whatever explains the assertability facts for disjunctions also explains the assertability facts for conditionals.

(3) I'm not sure about — disjunctions of conditionals with different antecedents are just really hard to process.

(4) is tricky - speeches of the form 'P and if not P then Q' are weird; there's a sense that something is being taken back. But Bennett has a case that really does seem problematic for Grice, namely the contrast between 'If P, not Q' and 'If Q, not P'.